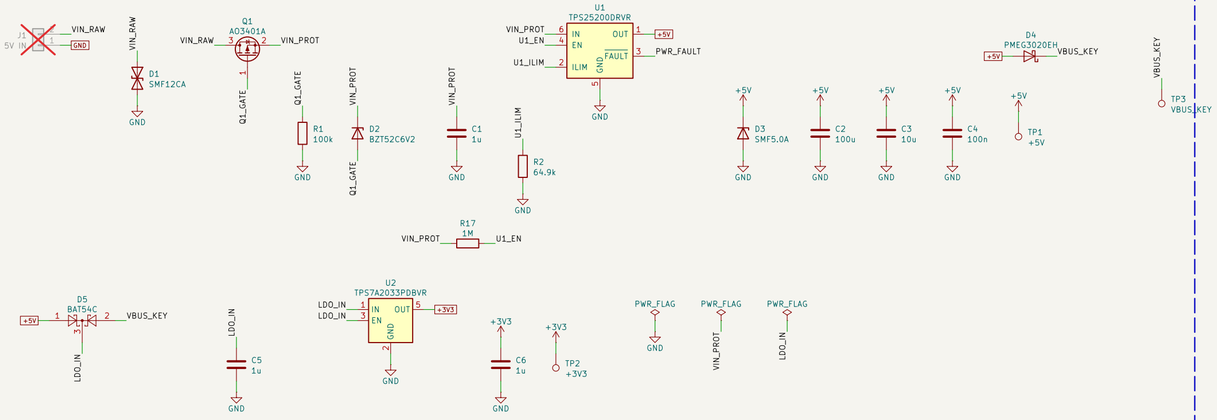
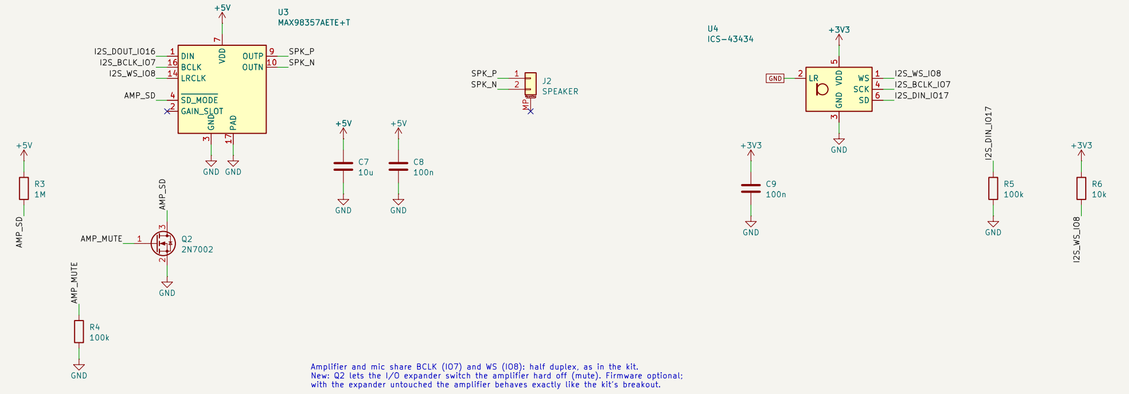
1 Power: 5 V terminal -> protection -> +5V; VBUS for the key; +3V3



Flow: terminal -> TVS -> reverse-polarity FET Q1 -> eFuse U1 (1.5 A limit, 5.4 V clamp, soft start) -> +5V node (TVS D3, bulk C2) -> amplifier, radar, glow ring. +5V -> Schottky D4 -> VBUS_KEY -> FFC -> touch board VBUS. USB-C on the touch board stays usable: D4 keeps USB power out of the hub; D5 runs the 3.3 V regulator from either side, so I2C, mic and the IO8 pull-up are live whenever the key has power.

2 Audio: I2S amplifier + speaker, I2S microphone (one shared I2S bus)

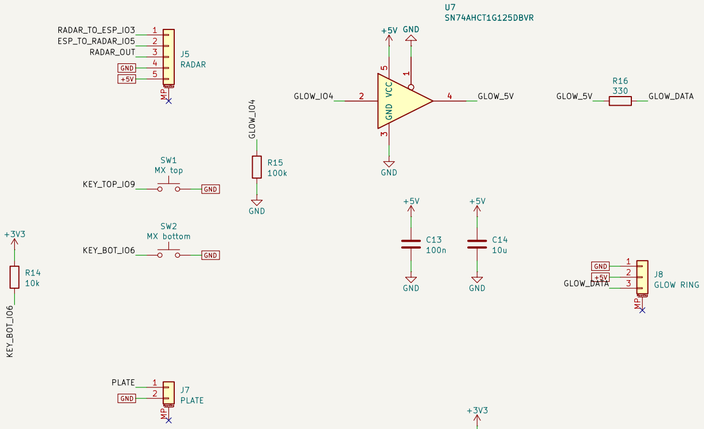


Amplifier and mic share BCLK (IO7) and WS (IO8): half duplex, as in the kit.
New: Q2 lets the I/O expander switch the amplifier hard off (mute). Firmware optional; with the expander untouched the amplifier behaves exactly like the kit's breakout.

3 I2C: I/O expander (plate, mute, radar OUT, power fault); screw holes

I2C addresses: TCA9534 0x20; VEML6030 0x10 and SHT4x 0x44 on the sensor tail, touch AXS5106L 0x63 and IMU QMI8658 0x6B (both on the touch board). No clashes.
Expander: P0 plate (in, low = pressed), P1 amp mute (out, high = mute), P2 radar OUT (in), P3 eFuse fault (in, low = fault), P4–P7 spare (firmware: outputs, low).

4 Connectors: FFC to the key, radar, keys, plate, glow ring, SHT tail



Key wires J4 (2 x 8 holes, 1.27 mm; same order as a later FFC):
1 VBUS_KEY
2 VBUS_KEY
3 GND
4 KEY_TOP_IO9
5 KEY_BOT_IO6
6 GLOW_IO4
7 ESP_TO_RADAR_IO5
8 RADAR_TO_ESP_IO3
9 I2C_SDA_IO18
10 I2C_SCL_IO19
11 GND
12 I2S_DOUT_IO16
13 I2S_BCLK_IO7
14 GND
15 I2S_WS_IO8
16 I2S_DIN_IO17



[TBD] marks open values; see README.md.
Generated by gen_schematics.py; the FFC pinout is shared with the pogo adapter.
5 V SELV only, from a certified supply in the NEIGHBOURING box. Never 230 V here.

aikos-home / RoomKey

Sheet: /
File: roomkey-hub.kicad_sch

Title: RoomKey hub (static, in the flush box)

Size: A2	Date: 2026-10-08	Rev: A (draft)
KiCad E.D.A. 10.0.7		Id: 1/1